

Original Article

One small step for citizens, one giant leap for science: *iNaturalist* records boost our understanding of the evolution of parental care in a clade of arachnids

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ABSTRACT

Citizen-science platforms have become powerful tools for documenting natural history, but their potential for reconstructing behavioural evolution remains largely unexplored. Here, we combine field and laboratory records with a systematic survey of *iNaturalist* observations to investigate the evolution of parental care in the superfamily Gonyleptoidea, the clade containing the most documented cases of care in harvestmen. We report 85 new records of parental behaviour (78 of maternal or paternal care and seven of no care), more than doubling the number of species with available data. These records, together with all published information, were mapped on to a newly assembled supertree of 165 species spanning nine families. Transition rates revealed that maternal care arises exclusively from ancestors without care, whereas paternal care evolves through two routes—from no care and from maternal care—suggesting distinct selective pressures underlying its repeated origins. Transitions from parental care back to no care

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are frequent in both sexes, suggesting that alternative egg-protection strategies (e.g. oviposition in cryptic sites and debris coverings) may compensate for the absence of parental attendance, while avoiding the costs of prolonged care. Our findings highlight the remarkable evolutionary lability of parental care in Gonyleptoidea and demonstrate that citizen-science data can substantially accelerate macroevolutionary research.

Keywords: citizen science; egg attendance; evolutionary transitions; harvestmen; Gonyleptoidea; maternal care; nesting; Opiliones; paternal care

INTRODUCTION

Harvestmen are the third most species-rich order of arachnids, with approximately 6900 species currently recognized (Kury et al. 2024). These species occur on all continents except Antarctica, but their highest diversity is concentrated in tropical regions characterized by warm and humid climates (Curtis and Machado 2007). Although there are some records of obligate or facultative parthenogenesis in harvestmen, most species reproduce sexually (reviewed in: Machado and Burns 2024). Among sexually reproducing species, fertilization is always internal. Sperm transfer occurs either indirectly, via a spermatophore in species of the sub-order Cyphophthalmi, or directly through a unique male intromittent organ in species belonging to the remaining extant suborders: Eupnoi, Dyspnoi, and Laniatores (reviewed in: Macías-Ordóñez et al. 2010). In general, two mating systems are recognized in harvestmen: (i) scramble competition, in which males actively search for receptive females, territories are not established, and aggressive interactions among males are typically restricted to the mate guarding period; and (ii) resource-defence polygyny, in which males establish territories at reproductive sites used by females and fight for control of these resources (reviewed in: Buzatto et al. 2013). Among all behavioural traits, parental care is arguably the most intensively investigated in harvestmen (reviewed in: Machado and Raimundo 2001, Machado and Macías-Ordóñez 2007, Requena et al. 2013, Machado and Burns 2024).

Parental care occurs in multiple harvestman lineages and encompasses both maternal and paternal egg attendance (*sensu* Smiseth et al. 2012). As in insects (Machado and Trumbo 2018), most documented cases of parental care in harvestmen involve maternal care (Machado and Raimundo 2001, Machado and Macías-Ordóñez 2007, Machado and Burns 2024). Although exclusive paternal care is rare across animals in general, it has evolved independently multiple times within harvestmen. Among arthropods, an extensive review indicates that paternal care has arisen independently in only 15 lineages, nine of which belong to the order Opiliones (Requena et al. 2013). This pattern is striking, as harvestmen represent only about 0.6% of described arthropod diversity yet account for 60% of the independent origins of paternal care in the phylum (Machado and Burns 2024).

Regardless of the caregiving sex, experimental evidence from numerous harvestman species demonstrates that egg attendance substantially increases offspring survival, primarily by preventing attacks from predators (maternal care: Machado and Oliveira 1998, 2002, Buzatto et al. 2007, García-Hernández and Machado 2018a; paternal care: Mora 1990, Requena et al. 2009, Quesada-Hidalgo et al. 2019). Moreover, most studies on parental care to date have focused on species within the superfamily Gonyleptoidea, which comprises nearly 2100 species and represents approximately 30% of all harvestman diversity (Kury et al. 2024). The multiple independent origins of both maternal and paternal care within Gonyleptoidea, combined with recent

phylogenetic advances (e.g. Kury 2014, Pinto-da-Rocha et al. 2014, Benavides et al. 2021), position this clade as a powerful system for investigating the macroevolution of parental care and the ecological conditions under which different caregiving strategies arise and are maintained (e.g. Machado et al. 2016, Santos et al. 2017).

An important yet under-utilized source of behavioural information on harvestmen is online photographic databases, such as *iNaturalist*. Despite the predominantly nocturnal and cryptic nature of harvestmen, the order has amassed more than 323000 observations as of April 2025. While most photos offer little or no behavioural information, an increasing number of records capture individuals feeding, aggregating, fighting, copulating, or engaging in parental care [i.e. when an individual is found on a clutch of eggs, as documented in several detailed studies; see examples in Machado and Burns (2024)]. With contributions from citizens and numerous taxonomists, many species are identified to the genus or species level. This makes the *iNaturalist* platform a valuable tool for rapidly accumulating information on the presence of parental care and the caregiving sex. This trend underscores the growing importance of citizen science in the 21st century (e.g. Mesaglio and Callaghan 2021, Roberts et al. 2022, Rosa et al. 2022, Rocha et al. 2024, Roger et al. 2024, Ozolina et al. 2025).

In this study, we present new records of the presence or absence of parental care for 85 harvestmen species, belonging to seven families within the superfamily Gonyleptoidea. These records were gathered over many years of field and laboratory work, conducted by different researchers across several Latin American countries, and were supplemented by a systematic search of the *iNaturalist* database. Using the extensive dataset gathered in this study, along with published data from the literature, we mapped the presence or absence of parental care, as well as the identity of the caregiving sex, on to a supertree of Gonyleptoidea, assembled here for the first time. By mapping the occurrence of parental care within this superfamily, we aim to uncover the evolutionary history of this behavioural trait in a clade that contains the majority of recorded cases of parental care in harvestmen (Machado and Burns 2024). Furthermore, our findings enable us to explore evolutionary transitions between presence and absence of parental care, and also the identity of the caregiving sex.

MATERIALS AND METHODS

Field and laboratory observations

The field and laboratory data for this study were collected by a large group of researchers across five Latin American countries: Brazil (18 species), Chile (eight species), Colombia (three species), Costa Rica (one species), and Ecuador (two species). There was no standardized effort for data collection, and the records were made opportunistically when harvestmen were found ovipositing or near their eggs. Whenever a female was found laying an egg, or an individual (male or female) was found in a typical

parental care posture on a clutch, both the parent and the eggs were photographed. From these photographs, we recorded the species' identity, sex of the caregiving individual, and the type of oviposition substrate. In general, high-quality photographs are sufficient to allow reliable identification to genus and species level. Given the pronounced sexual dimorphism observed in most species of Gonyleptoidea (Pinto-da-Rocha and Giribet 2007), the sex of the caregiving individual can also be confidently determined from photographs. In some cases, however, voucher specimens were collected in addition to photographic records and deposited in the Museu Nacional, Rio de Janeiro [*Deltaspidium asperum* (Perty, 1833), *Geraecormobius bispinifrons* Roewer, 1943, *Goniosoma roridum* Perty, 1833, *Gonyleptellus angeloi* Ázara and Kury, 2019, *Mitobates elegantulus* (Mello-Leitão, 1940), *M. pulcher* (Mello-Leitão, 1932), *M. triangulus* Sundevall, 1833, *Obidosus longipalpis* (Roewer, 1943), *Quindina* sp.1 and sp.2, *Ubatubesia amplicoxae* Pinto-da-Rocha, 2000, and *Ventriovomer ancyrophorus* (Butler, 1873)] or in the Museu de Zoologia da Universidade de São Paulo, São Paulo [*Ampheres fuscopunctatus* (Soares, 1942), *Deltaspidium orguense* (Soares and Soares, 1954), *Liogonyleptoides inermis* (Mello-Leitão, 1922), *Neogonyleptes karschii* (Sørensen, 1902), *Pachylus paessleri* Roewer, 1913, and *Spinivunus adumbrates* Roewer, 1943], both in Brazil. These voucher specimens were examined to confirm the identifications derived from the photographic records, particularly when image quality was insufficient to ensure species identification.

We do not have detailed behavioural observations on how long parents remain on their eggs, except in cases of no care, where we observed females abandoning the eggs after oviposition. However, our focus here is solely on the presence or absence of parental care and the identification of the caregiving sex. In some laboratory observations, we documented isolated single eggs laid on the substrate (e.g. rocks or fallen logs), with no adult individual found nearby. We categorized these cases as instances of no care, as described for several Gonyleptoidea species (e.g. Juberthie 1972, Willemart 2001, Stanley 2011, Zatz et al. 2011).

Search in the iNaturalist database

iNaturalist is a global citizen-science platform that allows users to upload georeferenced observations of organisms, including photographs, sounds, and associated metadata. Observations are identified through a community-based process in which multiple users propose or refine taxonomic identifications. An observation reaches Research Grade status when it includes a date, geographic coordinates, supporting media, and when the community converges on a taxonomic identification—typically defined as a majority agreement at the species-level.

The iNaturalist database search was conducted on 20 and 21 April 2025, using the names of the 15 families currently recognized as part of Gonyleptoidea (Kury et al. 2024) as search terms. The search returned the following number of observations: Agoristenidae = 84, Ampycidae = 168, Askawachidae = 1, Cosmetidae = 9203, Cranidae = 2954, Cryptogeobiidae = 35, Gerdesiidae = 1, Gonyleptidae = 9748, Manaosbiidae = 193, Metasarcidae = 20, Nomoclastidae = 190, Otilioleptidae = 0, Prostygynidae = 7, Stygnidae = 693, and Stygnopsidae = 183. Additionally, we searched for all photos tagged with Gonyleptoidea ($N=24142$), as some of them were not identified to the family

level. While most of these photos in this search had already been included in our search for individual families, 662 were new and contributed to the discovery of additional records of parental care. In all cases, we verified the identification of the photos, adding the best identification possible when necessary. Identifications were independently assessed by co-authors with taxonomic expertise in the respective lineages of Gonyleptoidea.

For each family, we reviewed all photos for evidence of individuals near eggs. In potential cases of parental care, we recorded: (i) the type of oviposition substrate, (ii) the identity of the caregiving sex, and (iii) the locality. In two instances, females were photographed laying a single egg, which we classified as cases of no care. In one of these instances, involving *Libitoides sayi* (Simon, 1879) (Cosmetidae), previous literature indicates that the species does not exhibit parental care, with females abandoning isolated eggs shortly after oviposition (Goodnight 1958). We excluded from our results all records from the family Nomoclastidae where only nests without the caring individual were photographed ($N=22$), as well as photos of clutches from the family Gonyleptidae where the parental individual was absent ($N=2$). These records were excluded because, based solely on nest structure or egg morphology, reliable species' identification is not possible. To assign a taxonomic identity and confirm both the presence of parental care and the sex of the caregiving individual, an adult had to be photographed in the nest or in direct association with the nest or clutch.

Construction of the supertree

We constructed a supertree for the superfamily Gonyleptoidea using a five-step approach. First, we used the ultraconserved element (UCE) phylogeny from Derkarabetian et al. (2023) as a backbone to establish relationships among families. This decision was based on the fact that UCEs provide accurate information on deep phylogenetic relationships (Faircloth et al. 2012) and that this phylogeny currently offers the most comprehensive taxonomic coverage of Gonyleptoidea families published to date. Second, we sampled available DNA sequences from GenBank for the cytochrome *c* oxidase subunit I gene (COI), the ribosomal RNA genes 12S, 16S, 18S, and 28S, as well as the nuclear marker for the elongation factor *EF1a* and the histone *H3* genes for 396 species. Sequences were aligned using MAFFT (Katoh et al. 2019) and concatenated for phylogenetic inference using RAxML (Stamatakis 2014). Third, we calibrated the molecular tree using penalized likelihood (Sanderson 2002), as implemented in the R package *ape* (Paradis and Schliep 2019) function *chronos*, with the age of the crown node of Laniatores (98.17 Mya) informed by the fossil *Petrobunoides sharmai* Selden et al., 2016 (Selden et al. 2016). Fourth, we incorporated topologies derived from morphological, transcriptomic, and UCE datasets, as well as additional taxonomic characters, to build a supertree using the matrix representation with parsimony (MRP) method (Baum 1992, Ragan 1992) implemented in TNT (Goloboff and Morales 2023) and with topological constraints for monophyletic families and genera. Because the MRP approach does not preserve branch lengths of the backbone tree, we used the node ages of the calibrated molecular phylogeny as secondary calibration points following the congruification method (Eastman et al. 2013). Finally, we applied taxonomic imputation to add missing species and pruned the set of trees to include only species for which data on the presence or absence of

parental care were available. To account for phylogenetic uncertainty introduced during taxonomic imputation, we generated 100 alternative trees by randomly resolving polytomies and conducted all analyses and estimates for this set of trees.

Mapping of parental care

Our behavioural database incorporates all previously published records of presence or absence of parental care [see table 2 in Machado and Burns (2024)], as well as the new records presented in this study (Tables 1 and 2). The complete dataset is presented in Table S1 of Supporting information S1.

To map the evolution of parental care across the superfamily Gonyleptoidea, we first identified the best model of evolutionary transitions among three discrete states (no care, maternal care, and paternal care) by fitting k-state Markov process (Mk) models. The Mk framework describes the evolution of discrete traits as a continuous-time Markov process, in which character states transition among one another along branches of the phylogeny (Lewis 2001). We compared five models. The first three were conventional models of trait evolution (Fig. 1): (i) Equal Rates, which assumes that all transitions occur at the same rate; (ii) Symmetrical Rates, which allows different rates between pairs of states but constrains forward and reverse transitions to be equal; and (iii) All Rates Different, which allows each possible transition to have its own rate. The remaining two were custom models of trait evolution (Custom QMatrix), which allow user-specified transition-rate structures (Fig. 1): (iv) Custom Symmetrical Rates, which assigns one shared rate to transitions between no care and either parental care type, and a second distinct rate for transitions between maternal and paternal care; and (v) Custom Partitioned Rates, which assigns two rates for transitions involving the gain and loss of care, independent of parental care type, and two distinct rates for transitions between maternal and paternal care.

We used Akaike's information criterion corrected for small samples (AIC_c) to compare the five models for each tree in our set of 100 alternative trees and identify the model best supported by the data. We then performed a marginal ancestral state reconstruction to estimate the marginal probabilities of ancestral states for each node (Pagel 1999), as well as a stochastic character mapping (N = 10 000) to estimate the proportion of transitions between states (Huelsenbeck et al. 2003). All analytical steps were implemented using the R packages *corHMM* (Boyko and Beaulieu 2021) and *PHYTOOLS* (Revell 2024).

RESULTS

Field and laboratory data

From December 1996 to July 2024, we collected behavioural data on the presence or absence of parental care in 32 species across 26 genera spanning six families (Table 1). Parental care is taxonomically widespread and includes both maternal and paternal egg attendance.

Maternal care was recorded in *Osornogyndes tumifrons* Maury, 1993 (*incertae sedis*), three species of Cranidae (Fig. 2B–D), one species of Stygnidae, and multiple subfamilies of Gonyleptidae, including Cearininae, Goniosomatinae (Fig. 3A), Roeweriinae (Fig. 3B, C), and seven of eight Pachylinae species (Fig. 4A–D) (Table 1). Paternal care was observed in one species of Ampycidae

(Fig. 2A), two species of Nomoclastidae (Fig. 7A, B), and multiple subfamilies of Gonyleptidae, including Caelopyginae (Fig. 5A), Gonyassamiinae (Fig. 5B), Progonyleptoidellinae (Fig. 5D, E), and two of three Gonyleptinae species (Fig. 5C) (Table 1). Finally, absence of parental care was found in several gonyleptid species, including one Gonyleptinae, one Pachylinae, one Neopachylinae, and all three Mitobatinae species (Fig. 6A, B) (Table 1). In some of these non-caring species, we directly observed females in the field laying single eggs, covering them with debris, and subsequently leaving them unattended (*Gonyleptellus angeloi*, *Mitobates elegantulus*, *M. pulcher*, and *M. triangulus*). In other cases, we found isolated and unattended eggs inside terraria in which only a single harvestman species was maintained (*Krateromaspis dilatata* and *Ubatubesia amplicoxae*).

Oviposition sites exhibit significant variation among species, even within the same family or subfamily (Table 1). For example, in Cranidae (Fig. 2B–D) and Gonyleptidae (Figs. 3–6), the two families with the greatest number of species in our dataset, females lay eggs on tree trunks, rocks, fallen logs, riverside banks, or vegetation, depending on the species (Table 1). Nomoclastidae stands out as the only family in which males actively construct nests prior to oviposition. In the genus *Quindina*, males build suspended nests into which females subsequently deposit their eggs. In one species, males build globular nests made mainly of mud (Fig. 7A), whereas in another species (probably closely related to *Q. pendula* Lancheros et al., 2024), males construct disc-shaped nests composed primarily of plant debris (Fig. 7B). In both cases, the nests hang from vegetation or exposed rootlets in sheltered pockets along roadside banks, and the eggs are embedded in the nest floor, making them difficult to detect.

iNaturalist data

Our search in the *iNaturalist* platform returned 185 photos of harvestmen close to their eggs (Supporting information S2). Of these photos, 131 (71% of the total) identified the harvestmen to species level, 47 (25% of the total) identified them to genus level, and seven (4% of the total) lacked taxonomic identification (two of them even lacking family-level identification). At the time the data were collected, 62% of the photos had Research Grade status (Supporting information S2).

The photos identified to species or genus level include representatives of six families of Gonyleptoidea: Cosmetidae (nine species, 10 photos), Cranidae (23 species, 53 photos), Gonyleptidae (44 species, 79 photos), Manaosbiidae (one species, one photo), Nomoclastidae (15 species, 34 photos), and Stygnidae (one species, one photo) (Table 2). Of the 93 species recorded in the *iNaturalist* database, 53 (58%) represent new records of no care, maternal care, or paternal care. An additional 31 species (33%) already have published records of parental care (Table 2). The remaining eight species [*Ventrivomer ancyrophorus*, *Phareicranus giganteus* (Roewer, 1913), *Goniosoma roridum*, *Metagyndes pulchellus* (Loman, 1899), *Neogonyleptes karschii*, *Pachylus paessleri*, *Quindina* sp.2, and *Obidosus longipalpis*] are also represented in the field and laboratory observations presented in Table 1.

Data obtained from *iNaturalist* corroborate the broad taxonomic distribution of parental care in Gonyleptoidea and reinforce the diversity of care strategies across families (Table 2). Both

Table 1. List of species of the superfamily Gonyleptoidea for which information on the presence or absence of parental care was gathered based on the authors' field observations and laboratory work. Information on the oviposition site and the locality where each species was observed is also provided. Numbers in parentheses following the species name indicate the number of individuals observed. The symbol # denotes specimens that were collected and maintained in the laboratory, where behavioural data were obtained. [Figure S1A–D](#) is provided in [Supporting information S1](#).

Species	Type of care	Oviposition site	Locality (date)
Ampycidae			
<i>Glysterus</i> sp. (N = 1; Fig. 2A)	Paternal	Underside of a leaf	Monteverde, Puntarenas, Costa Rica (June 2015)
Cranidae: Cranainae			
<i>Neocranaus</i> sp. (N = 1; Fig. 2B)	Maternal	On a tree trunk	Manizales, Caldas, Colombia (September 2009)
<i>Phareicranaus giganteus</i> (N = 4; Fig. 2D)	Maternal	Riverside banks and rock walls at cave entrance	Mera, Tungurahua, Ecuador (March and July 2024)
<i>Ventrivomer ancyrophorus</i> (N = 1; Fig. 2C)	Maternal	Underside of a leaf	La Unión del Toachi, Pichincha, Ecuador (December 2021)
Gonyleptidae: Caelopyginae			
<i>Ampheres fuscopunctatus</i> (N = 2; Fig. 5A)	Paternal	Underside of leaves	Ubatuba, São Paulo, Brazil (January 2010)
Gonyleptidae: Cearininae			
<i>Liogonyleptoides inermis</i> (N = 2; Fig. S1A)	Maternal	Natural cavities in tree trunks and rocks	Araraquara, São Paulo, Brazil (November 2004)
Gonyleptidae: Goniosomatinae			
<i>Goniosoma roridum</i> (N > 30; Fig. 3A)	Maternal	Roadside banks and rocks	Nova Friburgo, Rio de Janeiro, Brazil (January 2024)
Gonyleptidae: Gonyassamiinae			
<i>Trichominua</i> sp. (N = 1; Fig. 5B)	Paternal	Bromeliad axil	Teresópolis, Rio de Janeiro, Brazil (September 2017)
Gonyleptidae: Gonyleptinae			
<i>Geraecormobius bispinifrons</i> (N = 3; Fig. S1B)	Paternal	Underside of leaves	Nova Friburgo, Rio de Janeiro, Brazil (January 2024)
<i>Geraecormobius</i> aff. <i>bispinifrons</i> (N = 1; Fig. 5C)	Paternal	Underside of a leaf	Teresópolis, Rio de Janeiro, Brazil (January 2010)
<i>Gonyleptellus angeloi</i> (N = 1)	No care	Roadside bank	Nova Friburgo, Rio de Janeiro, Brazil (May 2024)
Gonyleptidae: Mitobatinae			
<i>Mitobates elegantulus</i> (N = 1)	No care	On a rock	Ubatuba, São Paulo, Brazil (December 1996)
<i>Mitobates pulcher</i> (N = 1; Fig. 6A)	No care	On a rock	Nova Friburgo, Rio de Janeiro, Brazil (May 2024)
<i>Mitobates triangulus</i> (N = 1; Fig. 6A)	No care	On a rock	Rio de Janeiro, Rio de Janeiro, Brazil (March 2013)
Gonyleptidae: Neopachylinae			
<i>Krateromaspis dilatata</i> [#] (N = 5)	No care	On rocks and fallen logs	São Paulo, São Paulo, Brazil (September 2016)
Gonyleptidae: Pachylinae			
<i>Calcarogynodes calcar</i> (N = 1)	Maternal	Under a fallen log	El Patagual, Concepción, Chile (June 2020)
<i>Metagyndes pulchellus</i> (N = 3; Fig. 4A)	Maternal	Under fallen logs	Cayucupil, Nahuelbuta, Chile (June 2020)
<i>Metagyndes</i> sp. (N = 4; Fig. S1C)	Maternal	Under fallen logs	Osorno, Los Lagos, Chile (June 2016)
<i>Neogonyleptes karschii</i> (N = 1; Fig. 4B)	Maternal	Under a fallen log	Los Lagos, Osorno, Chile (January 2010)
<i>Pachylus paessleri</i> [#] (N = 1)	Maternal	Under a rock	Valparaíso, Valparaíso, Chile (March 2010)
<i>Sadocus asperatus</i> (N = 1; Fig. 4C)	Maternal	Inside a cavity in a tree trunk	Cayucupil, Nahuelbuta, Chile (February 2019)
<i>Spinivunus adumbratus</i> (N = 1; Fig. 4D)	Maternal	Under a fallen log	Los Lagos, Osorno, Chile (January 2010)
<i>Ubatubesia amplicoxae</i> [#] (N = 1)	No care	Rotten logs	Ubatuba, São Paulo, Brazil (April–May 2016)
Gonyleptidae: Progonyleptoidellinae			
<i>Deltaspidium asperum</i> (N = 10; Fig. 5E)	Paternal	Underside of leaves	Nova Friburgo, Rio de Janeiro, Brazil (January 2024)

(Continued)

Table 1. Continued.

Species	Type of care	Oviposition site	Locality (date)
<i>Deltaspidium orguense</i> (N = 1)	Paternal	Underside of a leaf	Teresópolis, Rio de Janeiro, Brazil (October 1999)
<i>Deltigalus</i> sp. (N = 1; Fig. 5D)	Paternal	Underside of a leaf	São Luiz do Paraitinga, São Paulo, Brazil (November 2004)
Gonyleptidae: Roeweriinae			
<i>Amazochroma carvalhoi</i> * (N = 1; Fig. 3B)	Maternal	Under a fallen log	Manaus, Amazonas, Brazil (October 2018)
<i>Roeweria virescens</i> (N = 1; Fig. 3C)	Maternal	Under a fallen log	Antonina, Paraná, Brazil (August 2008)
Nomoclastidae			
<i>Quindina</i> sp.1 (N > 30; Fig. 7A)	Paternal	Globular nests built by the male	San Antonio del Tequendama, Cundinamarca, Colombia (June 2024)
<i>Quindina</i> sp.2 (N = 3; Fig. 7B)	Paternal	Disc-shaped nests built by the male	San Antonio del Tequendama, Cundinamarca, Colombia (June 2024)
Stygnidae: Stygninae			
<i>Obidosus longipalpis</i> (N = 3)	Maternal	Inside unoccupied nests of the bird <i>Momotus</i> sp. in roadside banks	Manaus, Amazonas, Brazil (November 2019)
Incertae sedis			
<i>Osornogyndes tumifrons</i> (N = 1; Fig. S1D)	Maternal*	Inside a cavity in a rotten log	Purranque, Los Lagos, Chile (January 2022)

*Both the female and male were found inside the cavity and close to the eggs.

maternal care and paternal care are widespread, although their frequency varies among clades. Maternal care predominates in Cranaidae (all eight genera) and is also present in Cosmetidae (two species), Stygnidae (*Obidosus longipalpis*), and multiple subfamilies of Gonyleptidae, including Bourguyiinae, Cearininae, Goniosomatinae, Gonyleptinae, Neopachylinae, Pachylinae, and Roeweriinae (Table 2). Paternal care is observed in Cosmetidae (five species), Manaosbiidae (*Bugabitia akini* Townsend et al., 2011), Nomoclastidae (all three genera), and multiple subfamilies of Gonyleptidae, including Caelopyginae, Gonyleptinae, Heteropachylinae, and Progonyleptoidellinae (Table 2). Finally, absence of parental care was recorded in two species of Cosmetidae (Table 2).

Oviposition sites also vary considerably across taxa, and detailed information is provided in Table 2. Consistent with the behavioural data collected in the field (Table 1), photographs from iNaturalist confirm that Nomoclastidae are distinctive in that males actively construct nests prior to oviposition. The structure of these nests can be broadly classified into three types (Table 2): (i) carpet-shaped nests (*Callcosma* sp.1 and sp.2, *Kichua rheimsae* Pinto-da-Rocha and Bragagnolo, 2017, *Quindina horologium* Betancourt-Caicedo et al., 2024, *Q. sanantonio* Betancourt-Caicedo et al., 2024, and *Quindina* sp.4, sp.5, and sp.6); (ii) arena-shaped nests [*Q. albomarginis* (Chamberlin, 1925) and *Q. limbata* (Roewer, 1943)]; and (iii) disc-shaped suspended nests (*Q. pendula*, *Quindina* sp.2, sp.3, and sp.7). A fourth type, a globular nest observed in *Quindina* sp.1, was recorded only in the field data (Table 1).

Mapping of parental care

In total, we gathered behavioural data for 165 species of the superfamily Gonyleptoidea, including 80 from the literature and 85 from the present study (Supporting information S1, Table S1). These species span nine families (plus one *incertae sedis* species): Ampycidae (one species), Cosmetidae (19 species), Cranaidae

(30 species), Cryptogeobiidae (one species), Gonyleptidae (87 species), Manaosbiidae (two species), Nomoclastidae (18 species), Stygnidae (six species), and Stygnopsidae (one species). Thirty species (18.2% of the total) exhibit no care, 87 exhibit maternal care (52.7% of the total), and 47 exhibit paternal care (28.5% of the total). Only one species, *Opisthoplatus proscipius* (Holmberg, 1876) (Gonyleptidae: Neopachylinae), exhibits flexibility in the form of parental care with a population from Uruguay exhibiting no care (Stanley 2011) and a population from Argentina exhibiting maternal care (Iglesias et al. 2022). In our mapping, we considered this species as exhibiting maternal care; however, all analytical results remain qualitatively unchanged if the species is instead considered as lacking care (Figs. S2, S3 and Table S2 in Supporting information S1).

The best model of trait evolution was the Custom Partitioned Rates, which includes four transition rates between states (Fig. 8): two rates for gaining and losing care (regardless of parental care type), and two additional rates for transitions between maternal and paternal care (Table S3 in Supporting information S1). Regarding the transitions between states, the state no care changed into both maternal and paternal care with high frequency (q2 in Fig. 8). Similarly, both maternal and paternal care reverted to no care with high frequency (q1 in Fig. 8). Transitions from maternal to paternal care were markedly less frequent (q4 in Fig. 8) and occurred primarily within the families Cosmetidae and Gonyleptidae (Fig. 9). In contrast, transitions from paternal to maternal care were estimated as absent (q3 in Fig. 8).

The supertree including all species from our dataset is shown in Figure 9, together with the mapped presence or absence of parental care and the identity of the caregiving sex. The mapping clearly indicates that the three states (i.e. no care, maternal care, and paternal care) evolved multiple times within the superfamily. According to our analysis, the ancestral state of Gonyleptoidea is ambiguous, with a mean marginal probability of 0.23 for no care, 0.25 for maternal

Table 2. List of species of harvestmen from the superfamily Gonyleptoidea for which we gathered information on the presence/absence of parental care from the *iNaturalist* database. After each species, the number of observations related to parental care is presented in parentheses. For each species, information is provided on the type of care (maternal, paternal, or no care), the oviposition site, and the country where the record was made. Species for which published records of parental care already exist in the published literature (see complete list in [Table S1 of Supporting Information S1](#)) are underlined. Information about the Research Grade and the precise location of each photo, as well as the link to each of them, is available in [Supporting information S2](#).

Species	Type of care	Oviposition site	Country
Cosmetidae: <i>incertae sedis</i>			
<i>Libitoides sayi</i> (N = 1)	No care	On the soil	United States
Cosmetidae: Cosmetinae			
<i>Paecilaema</i> sp. (N = 1)	No care	On a tree trunk	Brazil
Cosmetidae: Cynortinae			
<i>Cynorta bromeliacia</i> (N = 1)	Paternal	Bromeliad leaf	Honduras
<i>Cynorta</i> sp.1 (aff. <i>circumbrosa</i>)* (N = 1)	Paternal	On a leaf	Costa Rica
<i>Cynorta</i> sp.2 (aff. <i>circumbrosa</i>)* (N = 1)	Paternal	On a leaf	Costa Rica
<i>Cynorta</i> sp.3 (aff. <i>punctitergum</i>)* (N = 2)	Paternal	Inside tank bromeliads	Costa Rica
Cosmetidae: Metergininae			
<i>Erginulus subserialis</i> (N = 1)	Maternal	On a tree trunk	El Salvador
<i>Meterginus</i> sp. (N = 1)	Maternal	Under a fallen palm leaf	Honduras
Cosmetidae: Taitoinae			
<i>Eucynortella</i> sp. (N = 1)	Paternal	Underside of a leaf	Costa Rica
Cranidae: Cranainae			
<i>Cranaus</i> sp.1 (N = 1)	Maternal	Underside of a leaf	Ecuador
<i>Cranaus</i> sp.2 (N = 1)	Maternal	On a rock	Ecuador
<i>Holocranaus</i> sp.1 (aff. <i>calcar</i>)* (N = 4)	Maternal	Under a brick	Colombia
<i>Holocranaus</i> sp.2 (aff. <i>conspicuus</i>)* (N = 1)	Maternal	On a dead leaf	Ecuador
<i>Holocranaus</i> sp.3 (aff. <i>laevifrons</i>)* (N = 1)	Maternal	On a rock	Ecuador
<i>Metacranaus</i> sp. (N = 1)	Maternal	Underside of a leaf	Ecuador
<i>Phalangodus</i> sp. (N = 1)	Maternal	On a rock	Colombia
<i>Phareicranaus albigyratus</i> (N = 1)	Maternal	On a tree trunk	Ecuador
<i>Phareicranaus arthrocentricus</i> (N = 1)	Maternal	On a tree trunk	Ecuador
<i>Phareicranaus giganteus</i> (N = 1)	Maternal	On a rock	Ecuador
<i>Phareicranaus gracilis</i> (N = 1)	Maternal	On a tree trunk	Ecuador
<i>Phareicranaus hermosa</i> (N = 3)	Maternal	On rocks and roadside banks	Ecuador
<i>Phareicranaus lucifer</i> (N = 3)	Maternal	On rocks	Ecuador
<i>Phareicranaus</i> sp.1 (N = 2)	Maternal	On tree trunks and rocks	Ecuador
<i>Phareicranaus</i> sp.2 (N = 1)	Maternal	On a rock	Ecuador
<i>Phareicranaus</i> sp.3 (N = 1)	Maternal	On a tree trunk	Ecuador
<i>Ventrifurca caffeinica</i> (N = 1)	Maternal	Under a rock	Ecuador
<i>Ventripila</i> sp. (N = 1)	Maternal	On a tree trunk	Ecuador
<i>Ventrivomer ancyrophorus</i> (N = 15)	Maternal	Underside of leaves and on rocks	Ecuador
<i>Ventrivomer</i> sp.1 (N = 1)	Maternal	Underside of a leaf	Ecuador
<i>Ventrivomer</i> sp.2 (N = 2)	Maternal	Underside of leaves	Ecuador
<i>Ventrivomer</i> sp.3 (N = 1)	Maternal	Underside of a leaf	Ecuador
<i>Ventrivomer</i> sp.4 (N = 1)	Maternal	Underside of a leaf	Ecuador
Gonyleptidae: Bourguyiinae			
<i>Bourguya trochanteralis</i> (N = 1)	Maternal	Inside a bromeliad	Brazil
Gonyleptidae: Caelopyginae			
<i>Ampheres leucopheus</i> (N = 2)	Paternal	Underside of leaves	Brazil
Gonyleptidae: Cearininae			
<i>Liogonyleptoides tetracanthus</i> (N = 1)	Maternal	On a rock	Brazil
<i>Parapachyloides uncinatus</i> (N = 1)	Maternal	Under wood	Brazil
Gonyleptidae: Goniosomatinae			
<i>Acutisoma longipes</i> (N = 8)	Maternal	On rocks	Brazil
<i>Goniosoma capixaba</i> (N = 2)	Maternal	On rocks	Brazil
<i>Goniosoma roridum</i> (N = 2)	Maternal	On rocks	Brazil
<i>Goniosoma varium</i> (N = 1)	Maternal	On a rock	Brazil
<i>Heteromitobates discolor</i> (N = 3)	Maternal	Underside of a leaf and on a rock	Brazil
<i>Mitogniella indistincta</i> (N = 2)	Maternal	On a rock and a cave wall	Brazil
<i>Serracutisoma catarina</i> (N = 5)	Maternal	On rocks	Brazil
<i>Serracutisoma fritzmulleri</i> (N = 1)	Maternal	On a rock	Brazil
<i>Serracutisoma gnaspini</i> (N = 1)	Maternal	On a rock	Brazil

(Continued)

Table 2. Continued.

Species	Type of care	Oviposition site	Country
<i>Serracutisoma inerme</i> (N = 1)	Maternal	On a rock	Brazil
<i>Serracutisoma molle</i> (N = 1)	Maternal	On a rock	Brazil
<i>Serracutisoma proximum</i> (N = 3)	Maternal	Underside of leaves, on rocks and tree trunks	Brazil
<i>Serracutisoma pseudovarium</i> (N = 2)	Maternal	On tree trunks	Brazil
<i>Serracutisoma</i> sp. (N = 1)	Maternal	On a rock	Brazil
<i>Serracutisoma spelaeum</i> (N = 8)	Maternal	On cave walls	Brazil
<i>Serracutisoma thalassinum</i> (N = 1)	Maternal	On a tree trunk	Brazil
Gonyleptidae: Gonyleptinae			
<i>Acanthogonyleptes fulvigranulatus</i> (N = 1)	Paternal	On a roadside bank (?)	Brazil
<i>Geraecormobius rohri</i> (N = 1)	Paternal	Inside a cavity in a tree trunk	Brazil
<i>Gonyleptes ater</i> (N = 1)	Paternal	Inside a cavity in a tree trunk	Brazil
<i>Juticus</i> sp. (N = 1)	Paternal	Inside a cavity in a roadside bank	Brazil
<i>Neosadocus bufo</i> (N = 2)	Maternal	Underside of leaves	Brazil
Gonyleptidae: Heteropachylinae			
<i>Heteropachylus inexpectabilis</i> (N = 1)	Paternal	Under a fallen log	Brazil
<i>Magnispina neptunus</i> (N = 1)	Paternal	Inside a cavity in a roadside bank	Brazil
Gonyleptidae: Neopachylinae			
<i>Opisthoplatus prospicuus</i> (N = 1)	Maternal	Under a fallen log	Brazil
Gonyleptidae: Pachylinae			
<i>Acanthopachylus robustus</i> (N = 4)	Maternal	Under rocks and a fallen log	Argentina
<i>Metagyndes martensii</i> (N = 1)	Maternal	Under a fallen log	Chile
<i>Metagyndes pulchellus</i> (N = 1)	Maternal	Under a fallen log	Chile
<i>Neogonyleptes chilensis</i> (N = 1)	Maternal	Under a fallen log	Chile
<i>Neogonyleptes karschii</i> (N = 1)	Maternal	Under a fallen log	Chile
<i>Neogonyleptes</i> sp. (cf. <i>docilis</i>)* (N = 1)	Maternal	Under a fallen log	Chile
<i>Pachylus paessleri</i> (N = 1)	Maternal	Under a rock	Chile
<i>Pachylus</i> sp. (cf. <i>chilensis</i>)* (N = 1)	Maternal	Under a fallen log	Chile
<i>Sadocus funestus</i> (N = 1)	Maternal	Inside cavity in a tree trunk	Chile
<i>Tumbesia</i> sp. (N = 1)	Maternal	Inside a cavity in the ground (?)	Chile
Gonyleptidae: Progonyleptoidelinae			
<i>Iguaepeia melanocephala</i> (N = 2)	Paternal	Underside of leaves	Brazil
<i>Iporangaia pustulosa</i> (N = 1)	Paternal	Underside of a leaf	Brazil
<i>Progonyleptoidellus striatus</i> (n = 3)	Paternal	Underside of leaves	Brazil
Gonyleptidae: Roeweriinae			
<i>Bunopachylus orientalis</i> (N = 1)	Maternal	Under a rock	Brazil
<i>Discocyrtanus oliverioi</i> (N = 2)	Maternal	Under fallen logs	Brazil
<i>Discocyrtanus pertenuis</i> (N = 1)	Maternal	Under a rock	Brazil
Manosbiidae			
<i>Bugabitia akini</i> (N = 1)	Paternal	Underside of a leaf	Panama
Nomoclastidae			
<i>Callosma</i> sp.1 (N = 1)	Paternal	Carpet-shaped nest on the ventral surface of a leaf	Colombia
<i>Callosma</i> sp.2 (N = 1)	Paternal	Carpet-shaped nest (not possible to determine the substrate)	Ecuador
<i>Kichua rheimsae</i> (N = 2)	Paternal	Carpet-shaped nest on the ventral surface of leaves	Ecuador
<i>Quindina albomarginis</i> (N = 13)	Paternal	Arena-shaped nest on tree trunks and fallen logs	Panama
<i>Quindina horologium</i> (N = 2)	Paternal	Carpet-shaped nest on the ventral surface of leaves	Colombia
<i>Quindina limbata</i> (N = 1)	Paternal	Arena-shaped nest on a fallen log	Costa Rica
<i>Quindina pendula</i> (N = 4)	Paternal	Disc-shaped nest hanging from the vegetation	Colombia
<i>Quindina sanantonio</i> (N = 1)	Paternal	Carpet-shaped nest on the ventral surface of a leaf	Colombia
<i>Quindina</i> sp.2** (N = 1)	Paternal	Disc-shaped nest hanging from the vegetation	Colombia
<i>Quindina</i> sp.3 (N = 1)	Paternal	Disc-shaped nest hanging from the vegetation	Colombia

(Continued)

Table 2. Continued.

Species	Type of care	Oviposition site	Country
<i>Quindina</i> sp.4 (N = 1)	Paternal	Carpet-shaped nest on the ventral surface of a leaf	Colombia
<i>Quindina</i> sp.5 (N = 1)	Paternal	Carpet-shaped nest on the ventral surface of a leaf	Colombia
<i>Quindina</i> sp.6 (N = 3)	Paternal	Carpet-shaped nest on the ventral surface of leaves	Ecuador
<i>Quindina</i> sp.7 (N = 2)	Paternal	Disc-shaped nest hanging from the vegetation	Ecuador
Stygnidae: Stygninae			
<i>Obidosus longipalpis</i> (N = 1)	Maternal	Dry leaf	Brazil

^aThe *iNaturalist* database does not allow the inclusion of this type of identification. Only the genus or the full species name is accepted. We included a more detailed identification in this table to make it more informative.

^bThis is the same *Quindina* sp.2 included in Table 1.

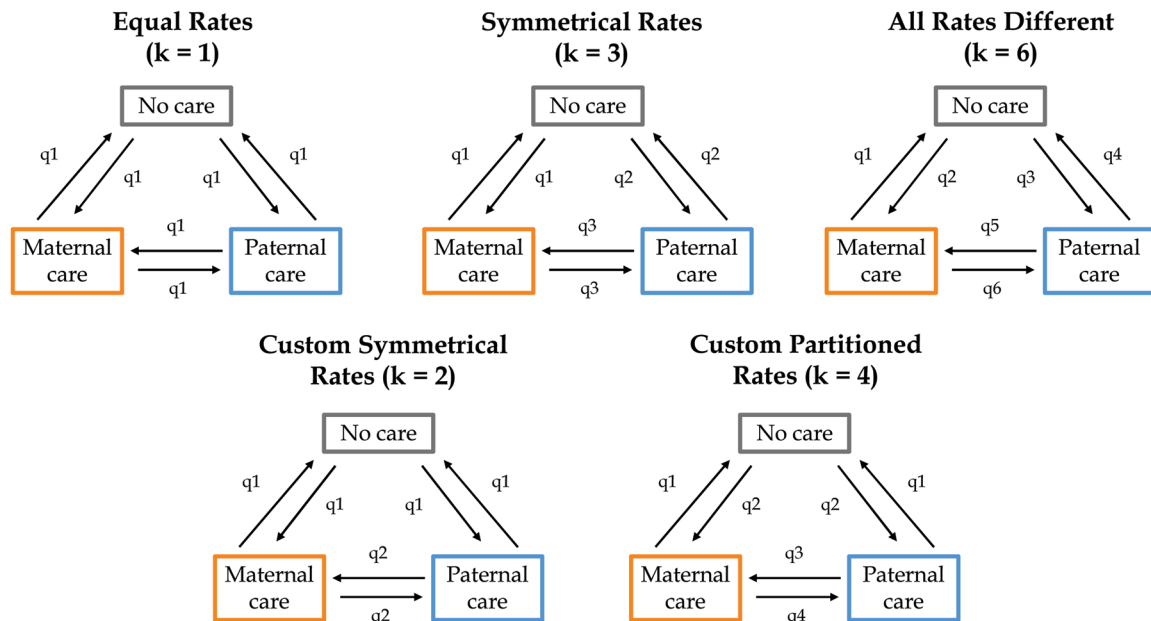


Figure 1. Models of evolution for the three-state trait care: no care, maternal care, and paternal care. In each model, states can transition between one another at different rates (q) along the phylogeny. The letter k indicates the number of distinct transition rates allowed in each model.

care, and 0.52 for paternal care. Table 3 shows the mean marginal probabilities associated with the ancestral state for each family represented by more than one species (i.e. excluding Ampycidae, Cryptogeobiidae, and Stygnopsidae). While no care is the most likely ancestral state for Cosmetidae, maternal care is most likely for Cranidae, and paternal care for Nomoclastidae (Table 3). For the remaining families, the probabilities are more evenly distributed among two or three states (Table 3).

DISCUSSION

Citizen science and the great leap for scientists

Between 1936 and 2025, the scientific literature documented behavioural data on the presence or absence of parental care for 80 harvestman species in the superfamily Gonyleptoidea (Supporting information S1, Table S1). In this study, we more than doubled these records, adding 85 new records. Data from *iNaturalist* alone contributed 62 of these records, eight of which were also independently obtained through field and laboratory

research (Fig. 10). Considering that the *iNaturalist* platform was launched in 2008, the rate of behavioural data accumulation in this database far exceeds that observed in traditional scientific work (Fig. 10). For comparison, from 2008 to 2025, researchers from various countries gathered data on only 36 species, highlighting the slower pace of data accumulation through conventional research efforts (Fig. 10). In fact, the *iNaturalist* platform has significantly accelerated the accumulation of biological data across various fields, complementing traditional fieldwork (e.g. Mesaglio and Callaghan 2021, Roberts *et al.* 2022, Rosa *et al.* 2022, Rocha *et al.* 2024, Roger *et al.* 2024, Ozolina *et al.* 2025). This rapid data accumulation allows for broader geographic and taxonomic coverage, enabling researchers to investigate macroevolutionary patterns at an unprecedented rate. This underscores the growing importance of citizen science, as it facilitates large-scale natural history data collection, often outpacing traditional scientific publishing.

Sampling biases present an inherent challenge when studying parental care in harvestmen (and any other taxon). It is easier to

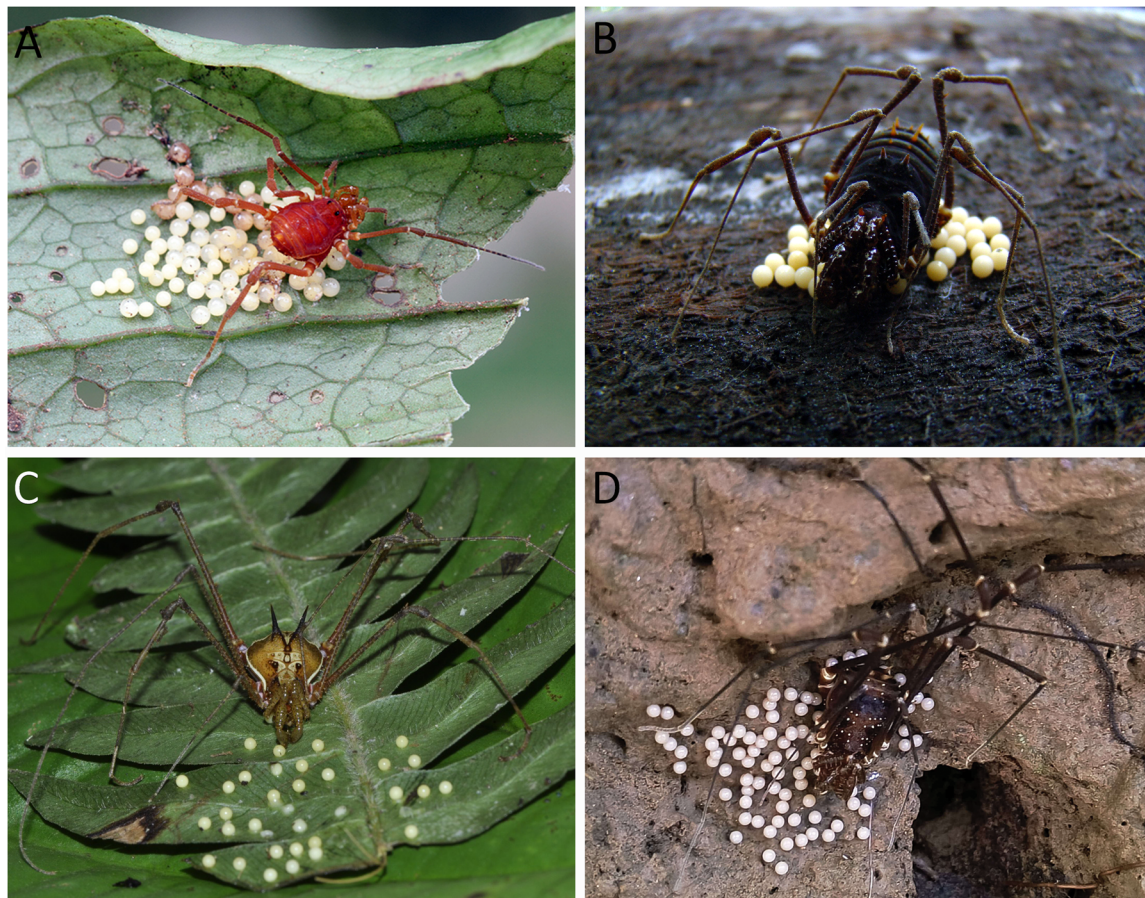


Figure 2. Cases of parental care in three families of Gonyleptoidea. A, male of *Glysterus* sp. (Ampycidae) caring for eggs laid on the underside of a leaf in Costa Rica (photo: Emilia Triana). Note the eggs at different stages of embryonic development, suggesting they were laid asynchronously, probably by more than one female. B, female of *Neocranaus* sp. (Cranaiidae) caring for eggs laid on a tree trunk in Colombia (photo: Luis E. Robledo-Ospina). C, female of *Ventrivomer ancrophorus* (Cranaiidae) caring for eggs laid on the underside of a fern in Ecuador (photo: Adriano B. Kury). D, female of *Phareicranaus giganteus* (Cranaiidae) caring for eggs laid on a riverside bank in Ecuador (photo: Joaquín Sáenz Mancheno).

document cases of parental care, whether maternal or paternal, as individuals engage in long-term interactions with their eggs (Machado and Raimundo 2001, Machado and Macías-Ordóñez 2007). In contrast, for species that do not exhibit parental care, finding isolated eggs in the field is extremely difficult. Even when such eggs are found, it is not possible to determine the species to which they belong unless the exact moment of oviposition is observed and an adult female is nearby (Fig. 6). This bias affects both researchers conducting fieldwork and citizen scientists capturing photographs. Indeed, of the 85 new records presented in this study, 78 correspond to instances of parental care, while only seven represent cases of no care. Among the no-care cases, five were obtained in the field (from researchers and citizens), and two were observed in the laboratory, where maintaining a single species in a terrarium allows for unambiguous assignment of any isolated eggs to the species in question. Therefore, it is important to recognize that, in our extensive effort to fill the gaps regarding the presence or absence of parental care in harvestmen, species exhibiting no care are likely to lag behind. This bias may impede a more detailed understanding of the transitions between no care and parental care within the group. Fortunately, the presence or absence of parental care in harvestmen of the superfamily

Gonyleptoidea appears to be phylogenetically conserved within genera, subfamilies, and some families (Fig. 9), allowing us to obtain a broader perspective even in the absence of data for many species.

Occasional observations of parental care and photos of harvestmen with their clutches provide valuable behavioural data, but it is important to recognize their limitations. In addition to lacking information on the duration and effectiveness of care against natural enemies, some observations show individuals of different sexes near the eggs (Supporting information S2). Could these be examples of biparental care? To date, no detailed, long-term behavioural study has documented biparental care in harvestmen. When care is maternal, females typically lay their eggs within a territory defended by a male (e.g. Machado and Oliveira 1998, Buzatto and Machado 2008, García-Hernández and Machado 2018a, Palaoro et al. 2022). In such cases, the male may engage in mate guarding after copulation, a behaviour that can last up to three days (e.g. Machado and Oliveira 1998, Buzatto and Machado 2008). During this period, both the male and the female are often observed near the clutch, but this does not mean that the male participates in egg attendance. Even after the mate-guarding period ends, the female may remate with the territorial male or

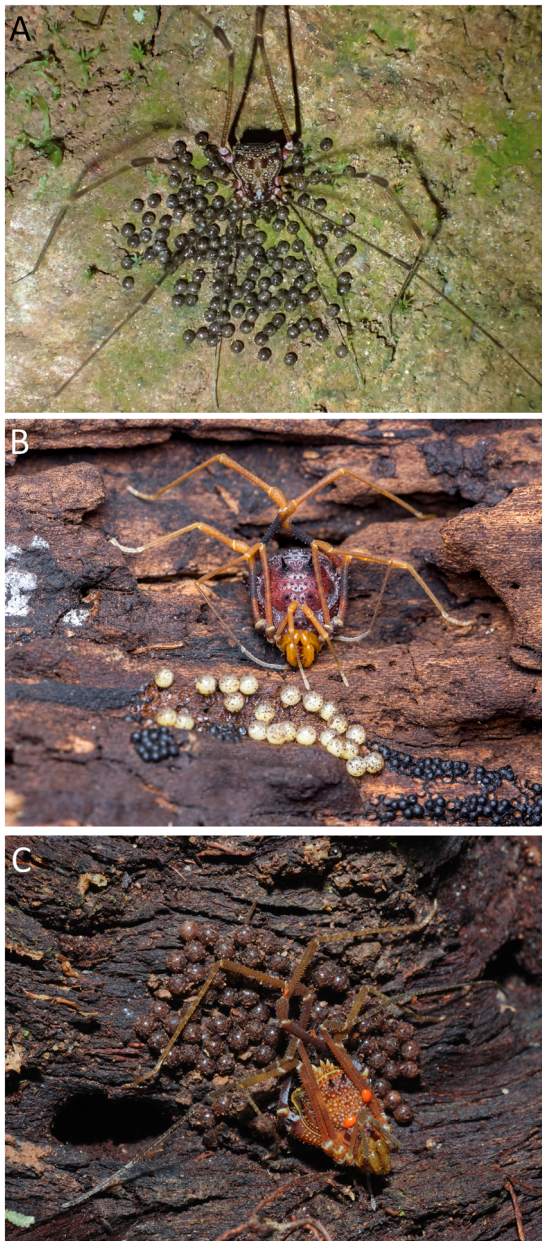


Figure 3. Cases of maternal care in the family Gonyleptidae. A, female of *Goniosoma roridum* (Goniosomatinae) caring for eggs on a roadside bank in Brazil (photo: Glauco Machado). B, female of *Amazochroma carvalhoi* (Roeweriinae) caring for eggs under a fallen log in Brazil (photo: John A. Uribe Rosas). C, female of *Roeweria virescens* (Roeweriinae) caring for eggs under a fallen log in Brazil (photo: Hubert Höfer).

other males that invade the territory (Buzatto *et al.* 2011, Muniz *et al.* 2015). As a result, there are multiple opportunities to observe males near egg-tending females in harvestmen. In fact, some photos of cranaids on *iNaturalist* show a pair near the clutch. A notable example is *Ventrivomer ancycrophorus*, a common species in Ecuadorian rainforests, for which we found 15 records on the platform and also collected data in the field. In this species, females are consistently observed near the eggs, while males are only occasionally recorded nearby ($N=2$). Moreover, a recent study suggesting the presence of biparental care in *Neocranaus* spp.

(Cranaidae) consistently photographed males near females caring for eggs in the early stages of embryonic development, which is consistent with mate-guarding behaviour [see fig. 14 in Villarreal *et al.* (2024)]. Therefore, we argue that the best explanation for males' proximity to the clutch in cranaids and some gonyleptids is related to mate guarding or remating attempts, rather than direct involvement in parental care.

When care is paternal, females may also occasionally be observed near the clutches. After copulation, females can remain for extended periods ovipositing or resting near the eggs in a quiescent position (Mora 1990, Nazareth and Machado 2010, Requena and Machado 2014a). In photos showing both the male and female near the clutches of these species, multiple lines of evidence can support the conclusion that the care is paternal rather than maternal. The most important of these is that, when males care for the offspring, eggs are laid asynchronously by different females, resulting in a clutch with eggs at various stages of embryonic development (Machado *et al.* 2004, Requena *et al.* 2013). This can be easily observed in photos (Figs 2A, 5C–E). However, it is important to note that in at least one harvestman species exhibiting maternal care, *Neosadocus bufo* (Mello-Leitão, 1923) (Gonyleptidae), females do not lay all their eggs synchronously, and the clutch also contains eggs at different stages of embryonic development (Chelini and Machado 2012). In this species, females temporarily abandon their clutches to protect themselves from adverse abiotic conditions or to forage (Chelini and Machado 2012). During temporary absences of the egg-tending female, the territorial male may occasionally be observed close to the clutch (G. Machado pers. obs.), a pattern consistent with amphisexual care—a distinct form of parental behaviour in which individuals of the non-caring sex may take over clutch guarding when the caring sex deserts (temporarily or permanently) or dies. Amphisexual care has been documented in other harvestman species, particularly within the subfamily Goniosomatinae (Gonyleptidae) (Machado and Oliveira 1998, Willemart and Gnaspini 2004, Buzatto and Machado 2009; but see Nolasco and Requena 2018). Interestingly, an *iNaturalist* photograph captured a male of *N. bufo* attending a clutch during a temporary absence of the female (Supporting information S2). The conclusion is that without detailed behavioural information on this species, this record could have been mistakenly interpreted as a case of paternal care.

Evolutionary history of parental care

Mapping the presence or absence of parental care, as well as the identity of the caregiving sex, on to the phylogeny of Gonyleptoidea revealed that the ancestral state is ambiguous (Fig. 9). In the family Nomoclastidae, however, paternal care is the most likely ancestral condition (Table 3). This clade provides an excellent opportunity to investigate the evolution of nest-building behaviour, a unique trait among all harvestmen [see discussion in Lancheros *et al.* (2024)]. Males of the genera *Kichua* and *Callcosma*, as well as some species of *Quindina* (e.g. *Q. sanantonio* and *Q. horologium*), construct a carpet-shaped nest on the ventral surface of leaves (Table 2; see also: Betancourt-Caicedo *et al.* 2024). According to our phylogeny, this nest type is probably the ancestral condition for the family (Fig. 9). There are three additional nest types in

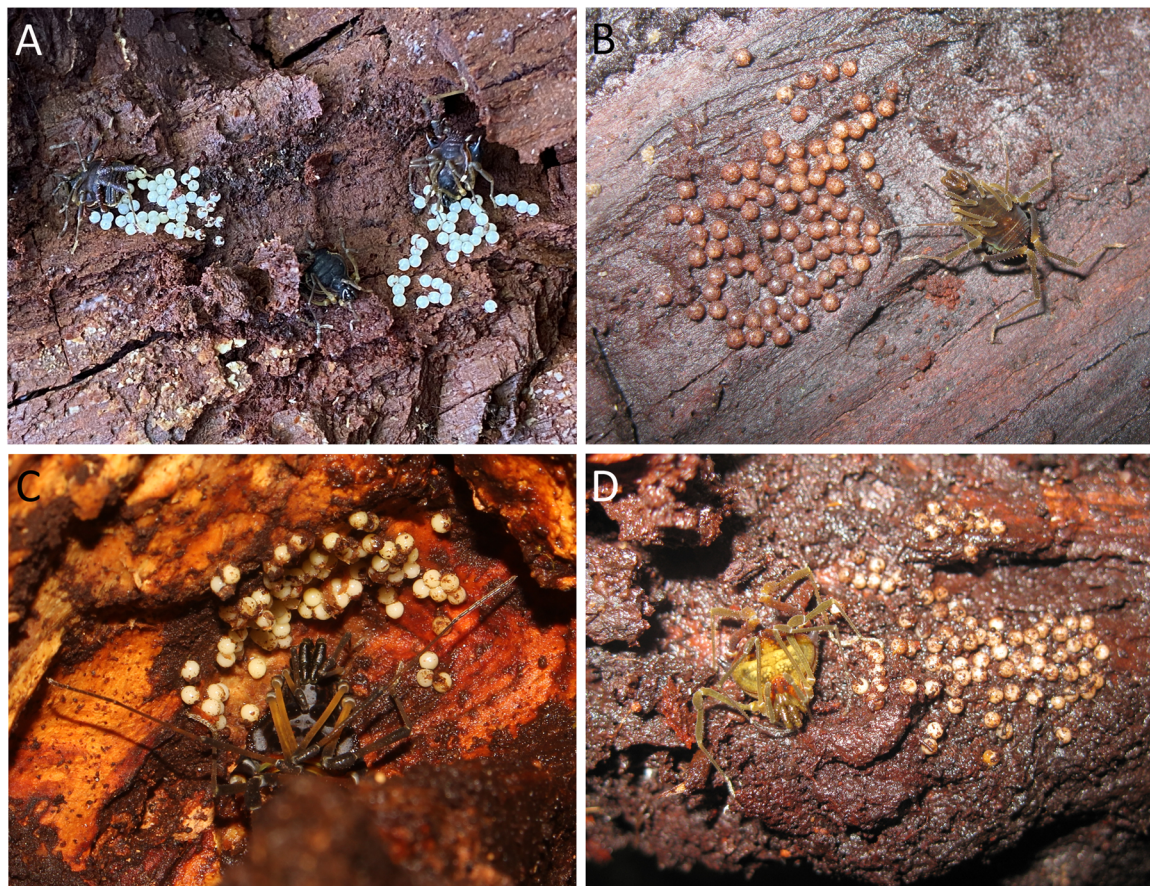


Figure 4. Cases of maternal care in the subfamily Pachylinae (Gonyleptidae). A, two females of *Metagyndes pulchellus* caring for the eggs under a fallen log in Chile (photo: Edgardo Flores). Note the male close to the eggs in the clutch on the right. B, female of *Neogonyleptes karschii* caring for the eggs under a fallen log in Chile (photo: Ricardo Pinto da Rocha). C, female of *Sadocus asperatus* caring for the eggs inside a natural cavity in a tree trunk in Chile (photo: Edgardo Flores). D, female of *Spinivunus adumbratus* caring for the eggs under a fallen log in Chile (photo: Ricardo Pinto da Rocha). In (B), (C), and (D), the eggs are covered by debris.

Quindina: (i) an arena-shaped nest built on fallen logs or tree trunks (Rodríguez and Guerrero 1976, Mora 1990, Pinto-da-Rocha and Bragagnolo 2017, Rojas et al. 2019); (ii) a disc-shaped nest hanging from vegetation (Lancheros et al. 2024; Fig. 7B); and (iii) a globular nest hanging from vegetation, described here for the first time (Fig. 7A). Unfortunately, the dataset includes many undescribed species whose phylogenetic placement is uncertain, preventing a detailed understanding of nest type evolution. Once these species are formally described and incorporated into the Nomoclastidae phylogeny, it will be possible to infer evolutionary transitions among nest types and to formulate hypotheses about the selective pressures that led to their diversification.

In the family Cranida, maternal care is the most likely ancestral state, and in all species for which behavioural data are available, females attend the eggs. This family, whose distribution is concentrated mostly in the Andean-Amazonian regions (Villarreal 2016), exhibits remarkable convergences in both morphology and behaviour with the subfamily Goniosomatinae (Gonyleptidae), whose species occur mostly along the Brazilian Atlantic Forest (DaSilva and Gnaspini 2009). Both clades include large-bodied species in which female egg attendance is the rule (Fig. 9). Primary oviposition sites include rocks, tree trunks, and leaves, depending on the species; however, females of some species, such as *Ventrivomer ancrophorus* (Cranida)

and *Serracutisoma proximum* (Mello-Leitão, 1922) (Goniosomatinae), can use all of these substrates in the same population (Tables 1, 2; see also: Buzatto et al. 2007). Moreover, the mating system of several species appears to be resource-defence polygyny, with males fighting for the possession of preferred oviposition sites (Cranida: e.g. García-Hernández and Machado 2018b, Palaoro et al. 2022; Goniosomatinae: Machado and Oliveira 1998, Buzatto and Machado 2008). As mentioned above, this mating system creates spatial proximity between territory-holding males and egg-tending females during the mate-guarding period, which may explain the frequent presence of males near females in cranids (Hunter et al. 2007, Villarreal et al. 2024), as well as male adoption of clutches whose mothers have died or deserted in goniosomatines (e.g. Machado and Oliveira 1998, Willemart and Gnaspini 2004, Buzatto and Machado 2009). Altogether, the convergence in morphology, parental care, oviposition-site use, and mating system suggest that cranids and goniosomatines may represent ecological equivalents (*sensu* Elton 1927) that evolved in distinct tropical biomes of South America.

In the families Cosmetidae and Gonyleptidae, the two most species-rich groups within the suborder Laniatores (Kury et al. 2024), there are species with and without parental care, and among those exhibiting care, both maternal and paternal

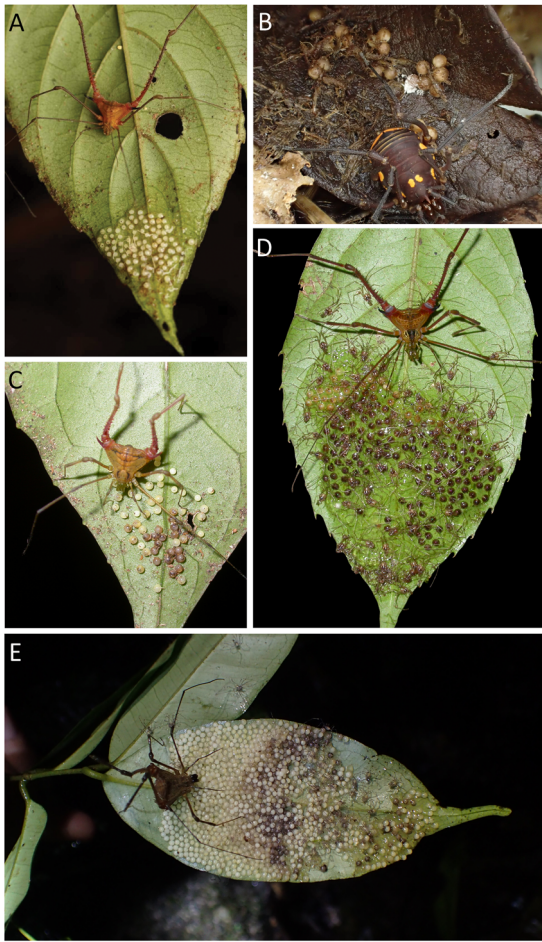


Figure 5. Cases of paternal care in the family Gonyleptidae. A, male of *Ampheres fuscopunctatus* (Caelopyginae) caring for eggs laid on the underside of a leaf in Brazil (photo: John A. Uribe Rosas). The clutch includes eggs and an early-hatched nymph. B, male of *Trichominua* sp. (Gonyassamiinae) caring for eggs laid on debris accumulated in a bromeliad axil in Brazil (photo: John A. Uribe Rosas). C, male of *Geraecormobius* aff. *bispinifrons* (Gonyleptinae) caring for eggs laid on the underside of a leaf in Brazil (photo: Bruno A. Buzatto). The clutch includes eggs in different stages of embryonic development. D, males of *Deltigalus* sp. and E, *Deltaspidium asperum* (both Progonyleptoidellinae) caring for eggs laid on the underside of a leaf in Brazil (photos: Glaucio Machado). The clutches include eggs in different stages of embryonic development and many early-hatched nymphs.

behaviours are present (Fig. 9). In Cosmetidae, no care appears to be the ancestral state (Table 3), with maternal care evolving independently once and paternal care evolving twice (Fig. 9). By contrast, the ancestral state in Gonyleptidae is ambiguous, although paternal care has a higher probability (Table 3). Paternal care occurs in all genera of the subfamily Heteropachylinae, in one genus of Gonyassamiinae (*Trichominua*), and it is probably the ancestral state in clade K92 (Fig. 9), which comprises six subfamilies (Kury *et al.* 2022). However, within clade K92, the subfamily Cearininae (represented here by three species in two genera) exhibits maternal care, and several species from different subfamilies appear to have independently lost parental care (Fig. 9). In the other major clade of Gonyleptidae—which includes Bourguiyinae, Goniosomatinae, Mitobatinae, Neopachylinae,

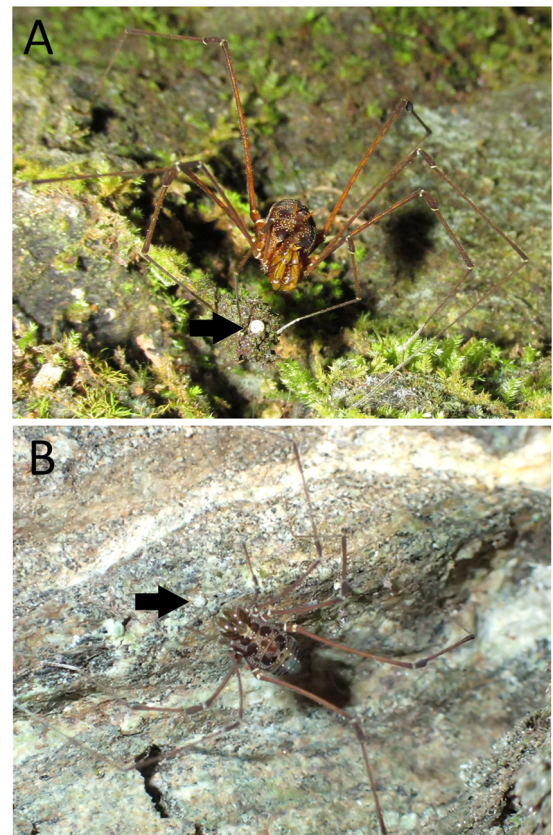


Figure 6. Cases of no care in the family Gonyleptidae. A, female of *Mitobates triangulus* and B, *M. pulcher* (both Mitobatinae) laying a single egg (indicated by the black arrow) on a rock in Brazil (photos: Osvaldo Villareal Manzanilla and Glaucio Machado, respectively). In both cases, the female abandoned the egg after oviposition and showed no parental care.

Pachylinae, Pachylospeleinae, and Roeweriinae—maternal care is widespread and represents the most likely ancestral state (Fig. 9). This condition has shifted to no care in a few isolated species and in all members of Mitobatinae (Fig. 9), whose females lay isolated eggs, cover them with debris, and subsequently abandon them (Willemart 2001, Zatz *et al.* 2011; Table 1; Fig. 6). Curiously, all Mitobatinae species included in our dataset occur in the same forested habitats where other species of this large gonyleptid clade exhibit maternal care. This suggests that the evolutionary loss of maternal care in Mitobatinae cannot be attributed to habitat differences and instead reflects intrinsic behavioural or life-history shifts unique to this lineage.

Our estimates of evolutionary transitions show that maternal care evolves exclusively from an ancestral state of no care (Fig. 8), a pattern consistent with what has been previously reported for insects (Gilbert and Manica 2015). Male care, in contrast, evolves from both no care and maternal care (Fig. 8), a markedly different pattern from insects, where paternal care has only originated from ancestors without care (Gilbert and Manica 2015). These two evolutionary routes probably correspond to distinct selective scenarios. The evolution of paternal care from an ancestral state of no care is compatible with the ‘overlapping broods’ hypothesis, which proposes that male care can emerge when its opportunity

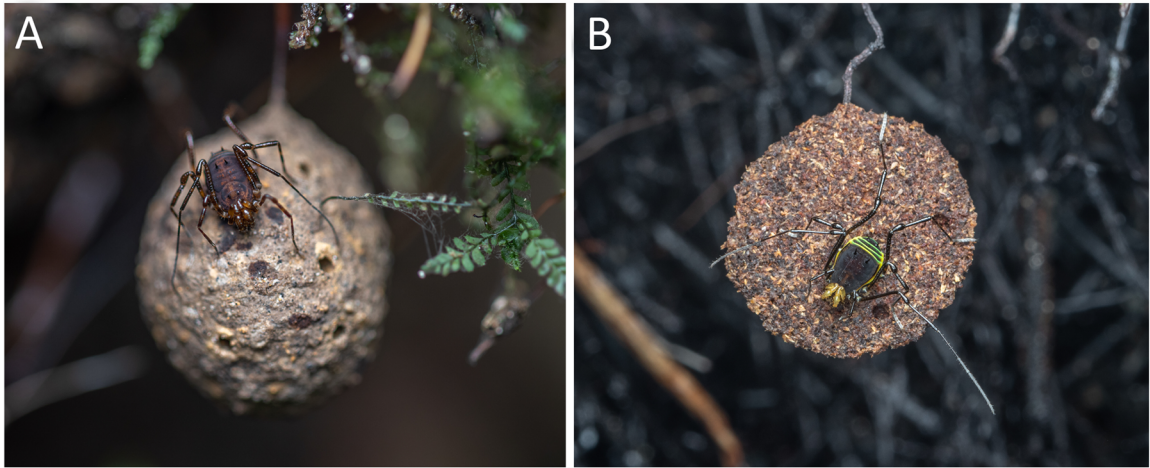


Figure 7. Cases of paternal care in the family Nomoclastidae. A, male of *Quindina* sp.1 caring for eggs laid on a globular nest hanging from a rootlet on a roadside bank in Colombia (photo: John A. Uribe Rosas). The nest is primarily composed of mud, and the eggs are embedded in the nest floor. The small holes in front of the male indicate places where eggs have hatched. B, male of *Quindina* sp.2 caring for eggs laid on a disc-shaped nest hanging from a fern rootlet on a roadside bank in Colombia (photo: John A. Uribe Rosas). The nest is primarily composed of dead plant matter, and the eggs are embedded in the nest floor. On the right side of the nest (around 5 o'clock), there is a small hole from which an egg has hatched.

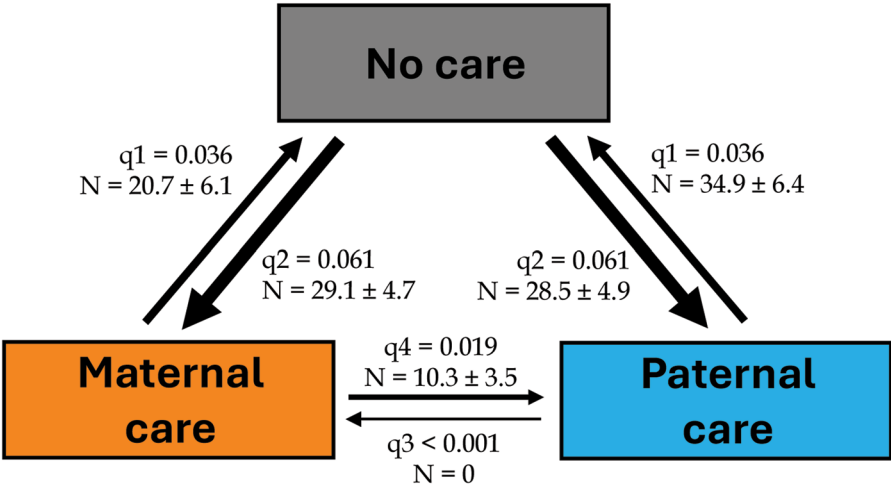


Figure 8. Transition rates among the three parental care states (no care, maternal care, and paternal care) in harvestmen of the superfamily Gonyleptoidea under the selected evolutionary model (Custom Partitioned Rates). The four transition rates are denoted by q , and arrow widths are proportional to their magnitude. The letter N indicates the average number of transitions between states ($\pm$ SD), estimated from 10 000 stochastic character mappings across 100 alternative topologies.

Table 3. Marginal probabilities of ancestral states for the presence and type of parental care in each family of Gonyleptoidea with more than one sampled species. Values are mean ($\pm$ SD) and were obtained from marginal ancestral state reconstruction based on 100 alternative topologies under the best-fitting model of trait evolution (i.e. Custom Partitioned Rates).

Family	Ancestral state		
	No care	Maternal care	Paternal care
Cosmetidae	0.9818 (0.0252)	0.0003 (0.0002)	0.0178 (0.0252)
Cranidae	0.0009 (0.0003)	0.9990 (0.0003)	0.0000 (0.0000)
Gonyleptidae	0.2202 (0.0029)	0.2324 (0.0032)	0.5473 (0.0045)
Manosbiidae	0.3169 (0.0030)	0.0478 (0.0010)	0.6353 (0.0023)
Nomoclastidae	0.0028 (0.0002)	0.0003 (< 0.0001)	0.9969 (0.0002)
Stygidae	0.4561 (0.0114)	0.3072 (0.0093)	0.2367 (0.0059)

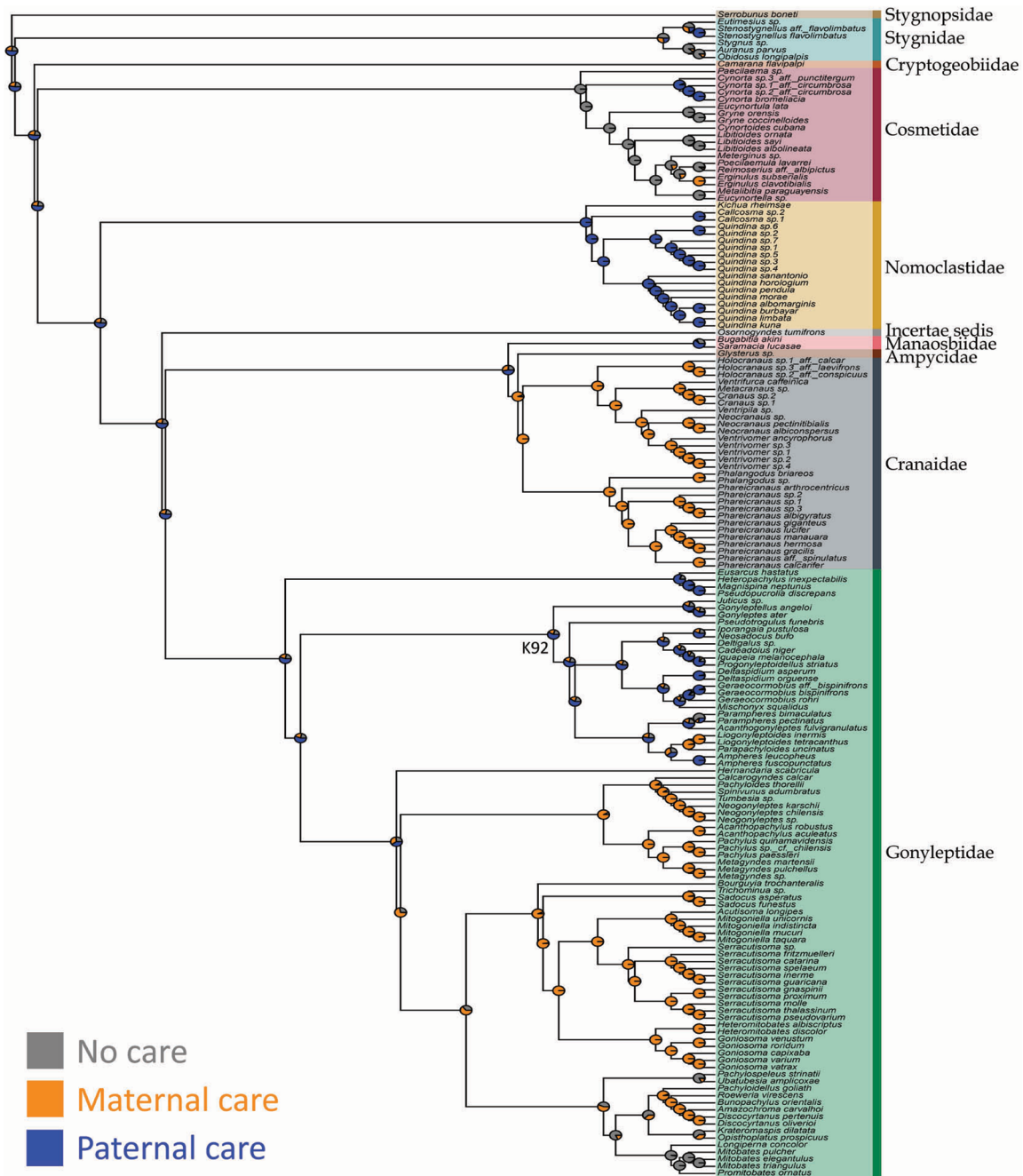


Figure 9. Marginal ancestral state reconstruction of parental care in the superfamily Gonyleptoidea, based on 100 phylogenetic trees. The phylogeny depicts relationships among the sampled species, with branches grouped by families (indicated by coloured bands on the right). K92 is a clade within the family Gonyleptidae that includes the subfamilies Caelopyginae, Cearinae, Gonyleptinae, Hernandariinae, Progonyleptoidellinae, and Sodreaninae. Circles at the nodes represent posterior probabilities of parental care states. The mapping reveals multiple independent origins of both maternal and paternal care across families, as well as several reversions to the absence of care. The topology shown is a randomly selected tree from the 100 fully resolved topologies used in the analyses. Branch lengths were rescaled for visualization purposes and do not represent evolutionary time.

costs—particularly the loss of mating opportunities—are reduced, such as when males are able to attend multiple broods simultaneously (Manica and Johnstone 2004). Under this scenario, selection may further favour paternal care if females evolve preferences for mating with caring males (Tallamy 2000). By contrast, the evolutionary transition from maternal to paternal care is compatible with the ‘enhanced fecundity’ hypothesis, which

proposes that paternal care may evolve when the reproductive costs of care are greater for females than for males, selecting for female desertion and leaving the brood in the sole care of the male (Maynard Smith 1977, Tallamy 2000). Overall, these contrasting routes highlight that the evolution of paternal care in harvestmen has not followed a single trajectory but instead reflects the interplay of different selective pressures across lineages.

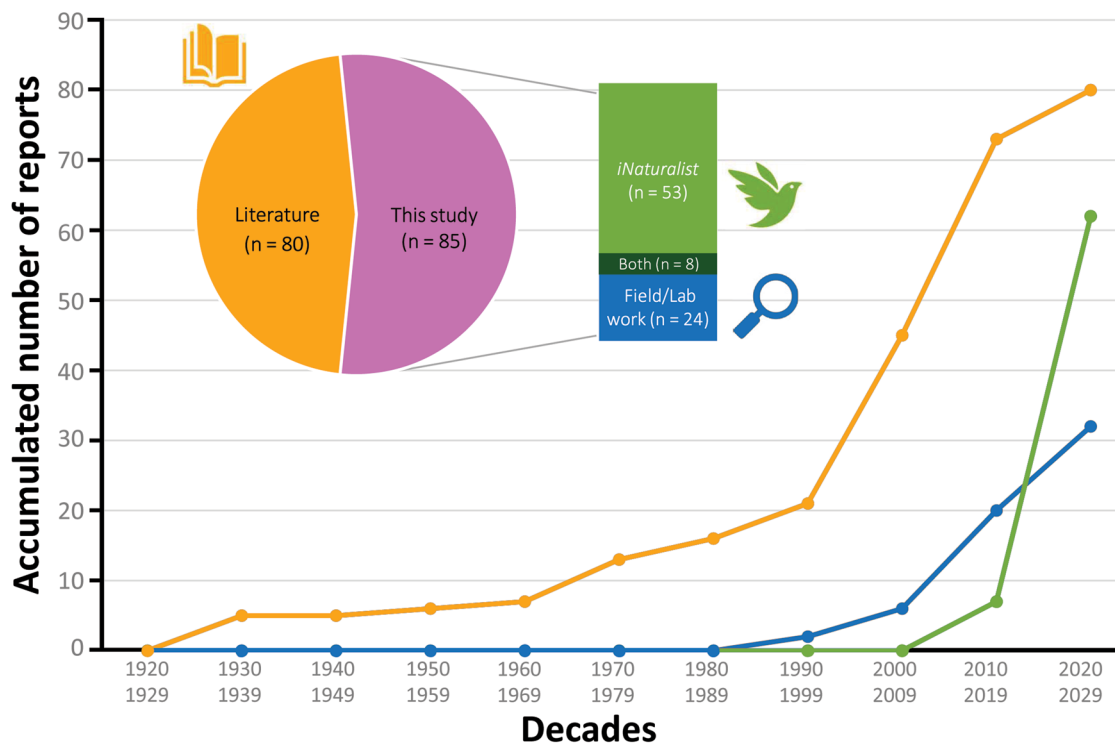


Figure 10. Accumulated number of reports on the presence or absence of parental care in the superfamily Gonyleptoidea from 1920 to the present. The orange line represents reports from the scientific literature, covering 80 harvestman species. The other two lines represent reports gathered in this study, covering 85 harvestman species. The blue line represents reports from field and laboratory observations, while the green line represents reports obtained from the *iNaturalist* database.

Egg attendance is the predominant form of parental care in harvestmen (Machado and Burns 2024, this study), and its costs are generally considered lower than those associated with other forms of care, such as offspring provisioning (Gilbert and Manica 2010; see also: Requena *et al.* 2012, Requena and Machado 2014b). Nevertheless, the frequent transitions from parental care—whether maternal or paternal—to no care across the phylogeny of Gonyleptoidea (Fig. 8) suggest that alternative forms of egg protection may also be effective in reducing predation. Such alternatives include laying eggs in cryptic microhabitats or covering clutches with debris (Machado and Raimundo 2001, Machado and Macías-Ordóñez 2007), which probably impose lower costs on parents, as they do not restrict foraging activities and, therefore, avoid potential negative effects on longevity and fecundity (Tallamy and Schaefer 1997; see also: Buzatto *et al.* 2007, Requena *et al.* 2012). Unfortunately, although several studies have evaluated the efficiency of parental care in enhancing egg survival in harvestmen [reviewed in: Santos *et al.* (2017)], to the best of our knowledge, no study has yet examined the protective role of alternative strategies. This knowledge gap limits our ability to understand why maternal care and paternal care are frequently lost in the group. However, we argue that in ecological contexts where alternative forms of egg protection effectively reduce predation risk, parental attendance may no longer provide sufficient benefits to outweigh its costs, thereby facilitating repeated evolutionary losses of care.

CONCLUSION

Our study demonstrates that citizen-science platforms, particularly *iNaturalist*, can substantially expand our empirical knowledge on parental care in harvestmen, yielding behavioural records at a pace unattainable through traditional field and laboratory work. Although sampling biases may favour the detection of parental care over its absence, the strong phylogenetic conservatism in this trait allows for robust inference of evolutionary transitions. These transitions indicate that maternal care generally arises from ancestors without care, whereas paternal care has evolved through multiple pathways, originating from both no care and maternal care. This pattern points to distinct selective forces shaping paternal care across lineages. The repeated loss of parental care, even in clades where maternal or paternal care is common, further suggests that alternative forms of egg protection can sometimes replace egg attendance. Behaviours such as oviposition in cryptic places and egg coating may offer sufficient defence against predators, while avoiding the foraging and longevity costs associated with prolonged egg attendance. The multiple independent gains and losses of parental care, combined with shifts in the identity of the caregiving sex, highlight harvestmen as an especially suitable model system for testing hypotheses on the evolution of parental care using a comparative framework. Finally, given that nearly 60% of the *iNaturalist* records represent new observations of no care, maternal care, or paternal care in harvestmen, we encourage scientists to further explore this platform as a source of novel biological data.

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AUTHOR CONTRIBUTIONS

G.M.: conceptualization, methodology, validation, investigation, data curation, writing—original draft, visualization. B.A.B.: investigation, writing—review and editing. D.S.C.: methodology, formal analysis, investigation, writing—review and editing. E.F.: investigation, writing—review and editing. S.G.H.: investigation, writing—review and editing. O.G.: investigation, writing—review and editing. L.A.G.: methodology, formal analysis, investigation, writing—review and editing. H.H.: investigation, writing—review and editing. A.B.K.: investigation, writing—review and editing. H.L.: investigation, writing—review and editing. M.M.: investigation, writing—review and editing. C.O.: investigation, writing—review and editing. J.P.S.: investigation, writing—review and editing. R.P.R.: investigation, writing—review and editing. L.E.R.O.: investigation, writing—review and editing. J.S.: investigation, writing—review and editing. D.S.B.: investigation, visualization, writing—review and editing. R.Q.H.: investigation, writing—review and editing. E.T.: investigation, writing—review and editing. O.V.: investigation, writing—review and editing.

SUPPLEMENTARY DATA

Supplementary data is available at *Zoological Journal of the Linnean Society* online.

CONFLICT OF INTEREST

The authors declare no conflict of interest.

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DATA AVAILABILITY

All behavioural data used in the study are available in the [Supporting information S2](#).

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